

HOW TO MAKE A GMO-FREE ZONE

Somebody draws a line on a map, writes down what may not be planted inside it, and gets that writing signed by whoever holds the power to sign. That is the whole of a zone. What follows is how each of the four instruments works, how to find out which one your jurisdiction leaves open to you, and what to do in the first month — in any country, starting with no money and no lawyer.

FOUR INSTRUMENTS · AT LEAST ONE IS OPEN WHERE YOU LIVE · THE ONE THAT WORKS IS RARELY THE ONE PEOPLE TRY FIRST

WHAT A ZONE ACTUALLY IS

A GMO-free zone is a defined area of land where engineered organisms are not planted, and where somebody has written down what happens if they are. The area can be one farm, a parish, a valley, a district, a province or a country. In Europe the GMO-free regions network counts more than 260 regions and over 4,500 municipalities and other local bodies that have declared themselves free of engineered crops in some form. Almost none started at a ministry.

The word declared carries the weight there. Four instruments travel under one name, signed by different people, failing in different ways.

THE FOUR INSTRUMENTS

1. A statute. An ordinance, by-law or prefectural regulation with legal force inside a boundary. Mendocino County, California, passed Measure H on 2 March 2004 with 57 per cent of the vote, the first county ban in the United States, after being outspent roughly six to one. Jackson County, Oregon, passed Measure 15-119 in May 2014 by two to one. Hokkaido enacted its regulation on the prevention of crossing through the cultivation of genetically modified crops in March 2005. Niigata requires permission to plant and backs it with penalties of up to a year's imprisonment or 500,000 yen. Eleven Japanese prefectures and three municipalities have gone this way. In the Philippines, Negros Occidental's provincial board passed Ordinance No. 07 of 2007, prohibiting the entry, planting, growing, selling and trading of engineered plants and animals across the province.

2. A coexistence rule. No prohibition, only conditions that make planting impractical: isolation distances, notice to neighbours, registration of field locations, liability for contamination. Hokkaido's distances are at least 300 m for rice, 1.2 km for maize and 2 km for sugar beet, about double the figures the national ministry set for its own research stations. The European Commission's coexistence guidelines (Recommendation 2010/C 200/01) go further, and tell member states to consider excluding cultivation from large

areas of territory where other measures cannot deliver sufficient purity.

3. A contract. Landowners bind themselves and one another by private agreement not to plant, in terms that run with the land or the lease. This is how much of the German and Austrian countryside became GMO-free while the legislative route was shut, and it is what still works when a higher government has taken the power to regulate away from your council.

4. A market standard. Organic certification, a non-GM supply specification, a processor's purchase contract, a school district's procurement clause. Part of it is already written: the US National Organic Program lists genetic engineering among its excluded methods at 7 CFR 205.105, and the EU organic regulation prohibits engineered material outright. Every certified farm inside your line already runs a private version of the rule you are asking for.

MOST WORKING ZONES ARE TWO OR THREE OF THESE AT ONCE

The statute is the part that gets photographed. The contracts and the buyers are usually what hold the ground. Build for both, and do not let the campaign stall while a council decides.

WHAT A ZONE CANNOT DO

Worth knowing before you spend a year on one. It does not stop imported grain, engineered feed or processed food reaching the shops, and it does not revoke a national approval. It usually does not bind land the drafters had no power over: Mendocino's ordinance leaves city, state, federal and tribal land free to plant. And it does not hold by itself. A zone that is not re-signed and re-walked stops being real some years before anyone stops describing it as real.

THE THREE ANSWERS THAT DECIDE YOUR ROUTE

Get these before you draft, because they decide which instrument is available and which is a wasted year.

Can your level of government regulate this at all? In the United States, mostly not. After Mendocino the state legislatures moved, and by 2017 a list kept by the American Seed Trade Association counted 29 states with seed-preemption laws. Oregon passed Senate Bill 863 on 3 October 2013 while a county petition was already circulating, preempting every county except Jackson, whose measure had already reached the ballot. On 18 November 2016 the Ninth Circuit held the Maui and Hawaii County cultivation bans preempted by the federal Plant Protection Act and by Hawaii's own scheme for harmful plants, and Kauai's disclosure ordinance preempted by state pesticide law. Three ordinances, gone at once. In the European Union the power sits with member states rather than regions: Upper Austria notified a regional ban, the Commission refused it in 2003, and the courts upheld that refusal in 2005 and on appeal in 2007. Directive (EU) 2015/412 later gave member states a lawful opt-out on grounds including land use, planning, socio-economic impact, coexistence and public policy. Seventeen used it across their whole territory, along with Wallonia, Scotland, Wales and Northern Ireland.

Is anything actually being planted, or applied for, near you? Search your national biosafety register and the Biosafety Clearing-House at bch.cbd.int for permits, trial sites and approvals in your district, and ask the authority directly in writing. A zone drawn around a real trial argues in facts. A zone drawn around nothing in particular is a statement of intent — legitimate, but easier to ignore.

Who owns the ground inside the line? The number of separate owners is the number of signatures the contractual route needs, and the publicly owned hectares are the ground you take first, because the council already holds the title and only has to decide what goes into the lease.

THE FIELD

everything a zone touches — what governs it, who signs it, what it is drawn around, and what crosses the line anyway

THE LAW ABOVE decides whether you are allowed to draw a line at all

NATIONAL BIOSAFETY LAW

Permits, trial conditions, registers, approvals. Cartagena Protocol behind it: 173 parties, decisions posted at bch.cbd.int.

PREEMPTION STATUTES

Which level of government is forbidden to regulate seed or planting. 29 US states by 2017. Oregon SB 863, 3 Oct 2013.

EU DIR 2001/18 + 2015/412

Cultivation opt-out held by member states, not regions. Grounds: land use, planning, socio-economics, coexistence, public policy.

REG (EU) 2026/1388

NGT plants split in two. Category 2 keeps the opt-out, coexistence and the label. Category 1 has none of them.

WHAT PUSHES BACK

PREEMPTION

One statute retires every local ordinance at once.

RIGHT TO FARM

Standing crops claimed as an established use of land.

REDEFINITION

Outside the definition, outside the instrument.

TRADE LAW

A restriction attacked as a barrier to commerce.

THE FOUR INSTRUMENTS a zone is whichever of these your jurisdiction leaves open to you

1 STATUTE

Ordinance, by-law or prefectural regulation, binding inside a mapped boundary. Mendocino 2004. Jackson County 2014. Hokkaido 2005. Negros Occidental Ord. No. 07, 2007.

2 COEXISTENCE RULE

No ban, only conditions: isolation distance, notice to neighbours, a public register of field locations, liability for loss. Hokkaido sets 300 m rice, 1.2 km maize, 2 km beet.

3 CONTRACT

A covenant among landowners, running with the land and the lease, enforceable by every other signatory. Reaches ground that preemption statutes do not.

4 MARKET STANDARD

Organic certification (7 CFR 205.105 in the US), non-GM purchase specifications, procurement clauses, tender terms on public land. No boundary; changes what is planted.

WHO SIGNS IT and who has to be in the room when they do

COUNCIL, COUNTY, PREFECTURE

A resolution states a position. An ordinance binds. A citizen initiative needs signatures — Jackson County, 4,668.

REGION OR MEMBER STATE

Where coexistence rules and cultivation opt-outs are set. Upper Austria showed a region cannot do it alone in the EU.

LANDOWNERS AND TENANTS

Parcel by parcel, by registry number, binding on successors. The owner count is the signature count.

BUYERS, CERTIFIERS, CANTEENS

Processors, exporters, organic certifiers, seed multipliers, school and hospital kitchens.

THE GROUND INSIDE THE LINE what the boundary is drawn around, and what each parcel stands to lose

1 THE BOUNDARY — DRAWN ON A PHYSICAL FEATURE AND FILED AS A MAP

ARABLE

The field the permit is about.

ORGANIC FARM

Certified. Loses the certificate first.

SEED MULTIPLICATION

Purity threshold written into the contract.

HIVES

Forage radius near 3 km. See Bablok, 2011.

WILD RELATIVES

One crossing, and it is permanent.

PUBLIC LAND

Council-owned. Let by tender, on your terms.

WATERCOURSE

Carries seed and residue downstream.

STORE AND MACHINERY

Where separation is actually lost.

WHAT CROSSES THE LINE ANYWAY the four routes a zone has to be maintained against, every season

POLLEN

Wind and insects, past whatever isolation distance the permit set.

SPILLED SEED

Truck routes, roadsides, rail sidings, field margins.

SHARED MACHINERY

One combine, two farms, no cleaning protocol.

A TENANT NOBODY TOLD

The parcel changed hands and the covenant was never passed on.

How to read this. The law your country already has sits across the top and presses down: it decides which of the four instruments is available to you, and the preemption box on the right closes the most doors. Each instrument is signed by the people directly beneath it, which is why the work runs upward from the ground rather than down from a ministry. The boundary at the bottom is drawn around parcels, each standing to lose something countable; those losses are the arguments a council reads. The red arrows are what crosses the line after the signing is done. Find the badge sitting where your own situation already is, and start at that step rather than at step one.

1 · DRAW THE BOUNDARY BEFORE YOU NAME IT

Get a parcel map from the cadastre, land registry, county assessor or municipal planning office, and the crop maps from the agricultural extension service if it keeps them. Draw a line that follows something a farmer can see and a lawyer can describe: a watershed, a valley floor, the service area of an irrigation district, a municipal boundary, the land inside a ring road. A boundary drawn on a physical feature survives challenge better than one drawn around the people who happen to agree with you, and it is also the boundary that matches how pollen and machinery actually move.

Then count what is inside it: parcels, distinct owners, hectares in public ownership, hectares under tenancy, and the crops grown. Mark the features that will become arguments or signatures later — hives within three kilometres of arable ground, certified organic land, seed-multiplication fields, wild relatives of any crop grown here, watercourses, the grain store, the machinery ring, the roads and rail sidings that spilled seed travels along.

2 · FIND OUT WHO IS ALLOWED TO SAY NO

Send one written question to your municipal clerk or legal service: does state, provincial or national law reserve the regulation of seeds, plants or agriculture to a higher level of government, and if so under which statute? Keep the reply. It decides whether you are drafting an ordinance or a covenant, and getting it wrong is how a year disappears.

Preempted. Ordinances are closed to you. Instruments 3 and 4 are not, and neither is your council's own land. Preemption statutes are written about regulating seed; they rarely reach a landowner's decision about their own field, a lease term, a purchase specification, or how a municipality disposes of ground it owns.

Shared. The common case. You cannot regulate seed, but land use, water, roads, procurement, agricultural extension and public land are yours, and a zone assembled from those powers reaches most of the same ground.

Open. Your council, prefecture or region can regulate cultivation directly. This is how Hokkaido and Niigata did it. It is rarer than people assume, and it is worth an hour to confirm rather than assuming either way.

FIVE QUESTIONS TO PUT IN WRITING, ONE PER LETTER

To the biosafety authority: is [organism] regulated under our biosafety law, and if it is not, under which instrument does the exemption sit?

To the ministry: have any release permits, field trials or approvals been notified for [district], and where is the register?

To the municipal clerk: is local regulation of seeds or plants preempted here, and by which statute?

To the region: do national or regional coexistence rules apply here, and what isolation distance do they set for each crop?

To the certifier: what buffer do you require around certified land, and what happens to certification if contamination is detected?

One closed question per letter, the deadline the law gives them for a reply, a copy to a second body, and a method that produces a receipt. Number the letters. Three unanswered ones are evidence.

THE PARCEL ROLL, AND WHAT TO WRITE DOWN WHILE THE MAP IS OPEN

Parcel or registry number. Owner. Tenant, if the land is let, and when the tenancy ends. Crop. Certification status. Distance to the nearest block of the same crop. Whether the parcel touches a watercourse, a public road, or the boundary itself.

Keep it where two people can open it and date every revision. This roll becomes the schedule attached to your instrument, the evidence the council reads, the buffer check the certifier wants and the register you re-walk each year. Assembling it is unglamorous, and it is why some zones survive a change of ownership.

3 · BUILD THE BODY BEFORE THE INSTRUMENT

A zone is signed by people who lose money if it does not exist. Find them first and the drafting gets easier, because they supply the numbers a council will actually read.

Farmers who save seed. Organic growers and the certifier who inspects them. Seed multipliers, who work to purity standards written into their contracts and lose the crop if they miss them.

Exporters and processors carrying non-GM specifications. Beekeepers. The irrigation district or water board. Whoever already convenes people locally — a co-operative, a union, a parish, a market association. One local scientist willing to be named.

Farmers first, and in the room. If the campaign looks like town people telling farmers what they may plant, it will fail, and it should. The people who own land inside your line belong inside the coalition rather than underneath it.

Ask each of them for a number, not an opinion. The organic grower: hectares certified, and the value of the premium. The seed multiplier: the purity threshold written into the contract and what happens if it is missed. The exporter: tonnage moving under a non-GM specification, and the buyer's name. The certifier: how many farms it inspects in this district. The beekeeper: hives, and how far they forage. Those five numbers fit on one page, and that page is what gets read into the minutes.

THE BEEKEEPERS ARE NOT BYSTANDERS

On 6 September 2011 the Court of Justice of the European Union decided *Bablok and Others v Freistaat Bayern* (C-442/09). Pollen from MON 810 maize grown on plots owned by the Bavarian state had reached hives between 500 metres and three kilometres away. The Court held that honey and pollen supplements containing that material could not be placed on the market without authorisation under the GM food regulation.

A beekeeper within pollen range does not merely object to the field next door. Their product becomes unsaleable, and unlike most neighbours they can prove the pathway. Get the hive locations onto your map early.

IF THE LAND IS LET, THE TENANT SIGNS TOO

In much of the world the person who decides what goes in the ground is the tenant, not the owner. A covenant signed only by owners leaves every let parcel open until somebody notices.

Ask for both signatures. Ask when each tenancy ends, and put the renewal dates in the parcel roll. That list of dates is your work programme for the next three years, and it is the single thing most often missing when a zone quietly stops covering a third of its own map.

4 · WRITE THE INSTRUMENT THAT MATCHES YOUR ANSWER

An ordinance. A definitions clause that names both the technique and the outcome, because definitions are where these instruments are most often defeated. A prohibition on cultivation, propagation and release inside the boundary. The boundary itself described by reference to a map filed with the clerk. Complaint-driven enforcement rather than an inspectorate, which is what made Mendocino affordable to a county that had no wish to fund one. A civil remedy — abatement, removal, injunction — rather than criminal penalties, which invite the argument that you are regulating in a field reserved to somebody else. A severability clause, so that losing one provision does not lose the instrument. A review date.

A coexistence rule. Here the numbers are the instrument. Isolation distance by crop; notice to neighbouring landholders a fixed number of days before planting; registration of every field location in a public register; record-keeping on shared machinery, transport and storage; and an allocation of liability for economic loss to the party that introduced the crop. Take your starting distances from national coexistence rules or from the seed-certification purity standards already used in your country, then justify anything longer by local conditions: narrow valleys, small islands, prevailing wind, hive density, the length of the growing season.

A contract. A covenant among landowners. Parcels identified by registry number, an obligation not to plant or to permit planting, terms binding on successors in title and on tenants, an initial term with automatic renewal, a right in every other signatory to enforce, and either liquidated damages or an agreed right to an injunction. Have a notary or solicitor read it once. That single reading is the only professional cost most zones ever incur.

A market standard. Get the clause into the documents that already exist: the school district's food contract, the hospital kitchen, the municipal canteen, the co-operative's purchase specification, the tender terms for grazing or arable leases on public land. Ask your certifier what buffer it requires around certified ground and draw that buffer on the map, because it is a distance somebody is already enforcing.

5 · GET IT ADOPTED

Find out which of three routes your council has. A resolution states a position, is not binding, and can be passed in weeks. An ordinance or by-law binds, moves slowly and goes through legal review. A citizen initiative puts it to a vote against a signature threshold — the Jackson County petition needed 4,668 signatures and its organisers collected about two thousand more than that.

Get on the agenda properly. Agenda deadlines usually close a fixed number of days before the meeting, and an item normally needs a sponsoring member or a qualifying petition. Bring the map, the parcel count and the money: organic hectares, seed contracts, export lines and certifications at risk are the figures that get read into the record. Bring farmers to the microphone rather than speaking for them. Ask for the vote to be recorded.

Expect a counter-delegation, and expect it to be funded. In Jackson County the opposition raised more than \$900,000, roughly \$400,000 of it from three firms, against the campaign's \$375,000, and lost two to one. Money buys mailers. It does not buy the neighbours.

6 · MAKE IT SURVIVE A LAWYER

Write the purpose clause in the language your legal system already accepts. In the European Union those grounds are set out in Directive (EU) 2015/412: town and country planning, land use, socio-economic impact, coexistence, agricultural policy, public policy. Track that vocabulary. Elsewhere the equivalent is your local government act's list of matters within municipal competence, and it is worth quoting it.

Do not claim to regulate food safety, plant pests or pesticide registration where a higher authority already occupies the ground. That reasoning is what ended the Hawaii ordinances: the counties framed their measures around plants that federal and state law already classified and controlled, and the framing was what the courts reached for.

7 · KEEP IT

A zone is a subscription rather than a purchase. Four things keep it alive. A register: who signed, which parcels, when each agreement renews, held by two unrelated people. An annual walk of the boundary, recording new tenancies, changes of ownership and changes of crop. A testing schedule: sample, date, photograph, and store the samples somewhere other than your own premises — contamination documented in the first season is worth far more than contamination noticed in year five. And a watch on the definition, because reclassification is the standing threat to every zone on earth.

Negros Occidental is the case to watch on this. Its 2007 ordinance held for eighteen years. In 2023 a 21-billion-peso feed mill and poultry project was proposed for the province; by September 2025 the provincial board was holding public hearings on an ordinance to replace the ban with a regulatory regime, and a coalition of farmers, certifiers, church leaders, scientists and student organisations was back in the same hall its predecessors had filled in 2007. A zone is repealed the way it was passed, by a vote in a room, and the vote arrives when an investment does.

8 · WHEN IT IS STRUCK DOWN, PREEMPTED OR IGNORED

Fall back one instrument rather than out of the fight. A council that may not ban planting county-wide can still write the prohibition into every lease of land it owns, into its procurement, and into the terms it attaches to grants and extension support. Landowners whose ordinance was struck down can sign the same prohibition to each other as a covenant, and no preemption statute touches it.

Then go up. Where cultivation power sits nationally, as it does across the European Union, a hundred municipal declarations are the material for a national opt-out, and that is precisely how the European network has been used: 64 regional governments inside it, and an opt-out list that eventually covered seventeen member states in full. Local declarations that individually bind almost nothing are, in aggregate, the evidence that a national government points at.

ARGUMENTS THAT SURVIVE CONTACT

Be ruthless about what you carry into a public meeting. The wrong argument does not merely fail. It discredits the people standing next to you, and gets quoted back at them for years.

THE ONE THAT LOSES

That eating it will poison you. The regulator has a folder of feeding studies and will produce it, the room stops listening, and every economic argument you brought afterwards is filed under the same heading. It also splits you from farmers, some of whom have grown these crops for a decade.

THE ONES THAT HOLD

Who carries the loss. A certificate, a contract or a premium is lost by someone who did nothing, and no document names who pays them.

The cost of coexistence. Somebody pays for buffers, separate machinery, separate storage and testing, and it is the non-GM grower. The EU-funded CO-EXTRA project estimated that holding maize seed below 0.3 per cent GM content could add more than 20 per cent to the cost of producing it.

Certification and the buffer. Organic standards already prohibit engineered material and already specify separation distances. You are asking for a rule that matches one your neighbours are inspected against.

Honey. Bablok, above. Nobody can fence bees, and a court has already said what happens to the honey.

Seed multiplication. Contract purity thresholds are numerical, audited and enforced, and a trial next door puts them at risk in a way that is easy to demonstrate.

Wild relatives and centres of origin. Where a crop's wild ancestors still grow, gene flow is not hypothetical and one crossing is permanent.

Coexistence is impossible here. Directive (EU) 2015/412 names high cost or impracticability of coexistence caused by geography — small islands, mountain zones — as a ground in its own right. Narrow valleys and fragmented holdings make the same case.

Seed ownership. Patents and plant-variety rights travel with the trait, and change what a farmer may lawfully do with the

harvest.

WHAT THEY DO BACK

They preempt. The most effective move available, because it does not require winning your argument. One state statute retires every local ordinance at once, and it is usually drafted while a campaign is still gathering signatures.

They invoke the right to farm. Existing growers argue that a prohibition takes an established use of their land. Anticipate it: exemption for standing crops, a transition period, a remedy aimed at future planting.

They redefine. Watch the definition as closely as you watch any application. Regulation (EU) 2026/1388, in force from 16 July 2026, splits plants made with new genomic techniques into two categories. Category 2 stays under the GMO rules, with member-state opt-outs and coexistence measures available. Category 1 is treated as equivalent to conventional breeding, carries no product label except on seed, and has no opt-out. Neither is permitted in organic production, though unavoidable traces of category 1 are tolerated. For a European zone that moves the ground under the instrument: the opt-out and the label both stop applying to the category most likely to arrive first. The implementing acts are being written now, which is when they can still be argued with.

They wait. These files move over five to fifteen years. Volunteer coalitions dissolve in three. Decide early who keeps the archive, and make sure two unrelated people hold the map, the roll and the signatures.

EIGHT WAYS PEOPLE LOSE THIS

1. Drafting an ordinance in a preempted jurisdiction, and learning that at the hearing.
2. Arguing food safety, and losing the room for the economic case that wins.
3. Drawing the boundary around the people who agree rather than a physical feature.
4. Writing criminal penalties into a local instrument, and handing over the preemption argument.
5. Passing it, then never re-signing, re-walking or re-mapping it.
6. Letting one person hold the roll, the map and the signatures.

7. Speaking for farmers instead of getting them to the microphone.

8. Going to the press before the parcels, owners and legal route are known.

WHERE TO LOOK, AND WHO ANSWERS STRANGERS

Documents. Your cadastre or land registry, for parcels and owners. Your national biosafety register and the Biosafety Clearing-House at bch.cbd.int, for what has been permitted near you. The EU GMO register and EFSA application register; APHIS and regulations.gov in the United States. And your own council's ordinance archive, because adapting an adopted text beats writing one.

People who answer strangers. The European GMO-free regions network at gmo-free-regions.org, which publishes country pages, declarations and the texts regions actually used. IFOAM Organics Europe, and your national organic certifier association, which loses money on contamination and answers quickly. Third World Network's Biosafety Information Centre. The African Centre for Biodiversity. GRAIN and No Patents on Seeds, on seed ownership. The Environmental Law Alliance Worldwide, which connects local lawyers in more than seventy countries and costs nothing to contact. Ask early; somebody has already drafted the instrument you need.

YOUR FIRST MONTH

Week 1. Parcel map. Boundary drawn on a physical feature. Count parcels, owners, public hectares, crops.

Week 2. Preemption question to the clerk, permit question to the biosafety authority. Both in writing. File the replies.

Week 3. The five who lose money — organic grower, seed multiplier, beekeeper, exporter, certifier. Meet them on their land.

Week 4. Pick the instrument your answers allow. Mark up an adopted text from elsewhere. Get onto an agenda, or get the first three signatures.

IF YOU DO ONLY ONE THING

Establish in writing whether your level of government is permitted to regulate this at all. Everything else in this manual is recoverable. A year spent drafting an ordinance in a jurisdiction where that power was taken away in 2013 is not.